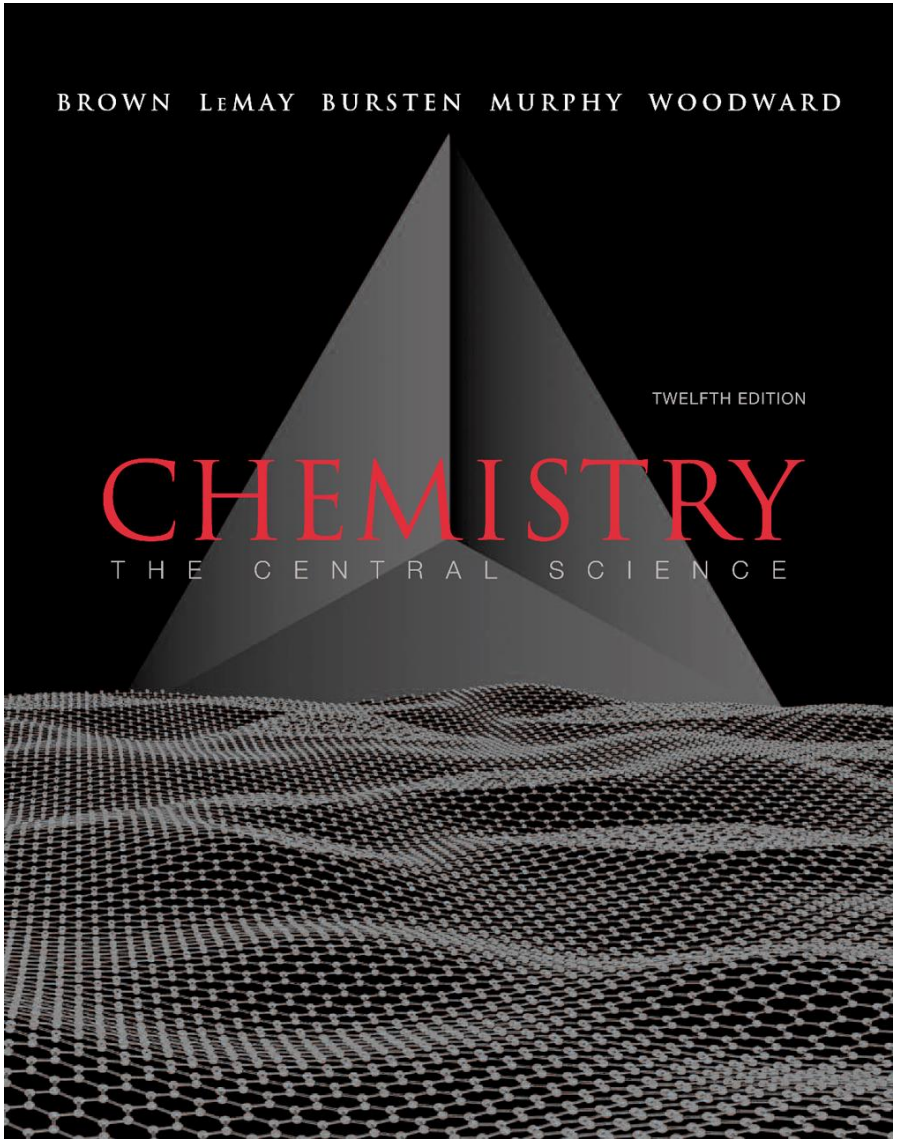


Lecture Presentation

Chapter 11

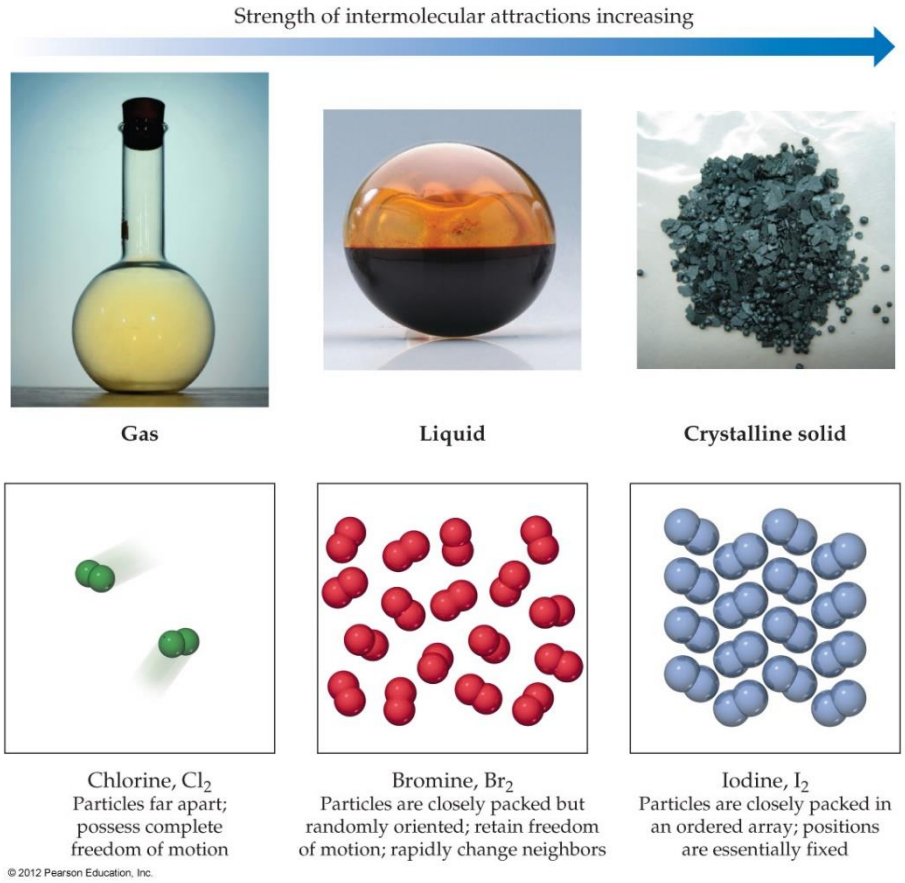
Liquids and Intermolecular Forces

John D. Bookstaver
St. Charles Community College
Cottleville, MO

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States of Matter

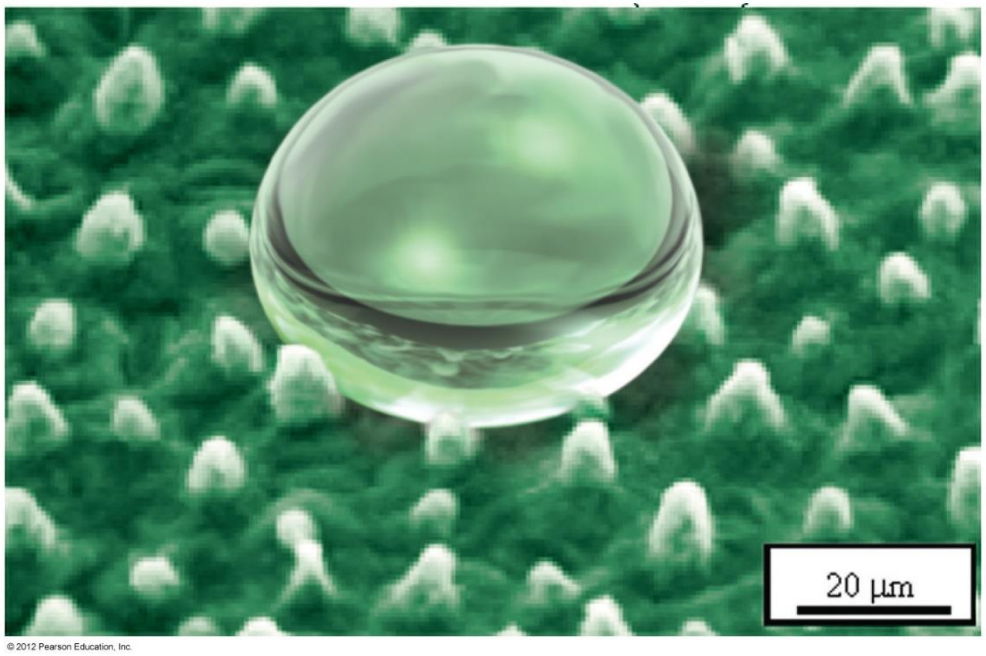
The fundamental difference between states of matter is the distance between particles.



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States of Matter

Because in the solid and liquid states particles are closer together, we refer to them as **condensed phases**.



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Intermolecular Forces

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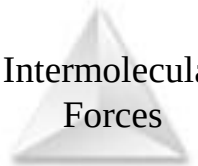
The States of Matter

TABLE 11.1 • Some Characteristic Properties of the States of Matter

Gas	Assumes both volume and shape of its container Expands to fill its container Is compressible Flows readily Diffusion within a gas occurs rapidly
Liquid	Assumes shape of portion of container it occupies Does not expand to fill its container Is virtually incompressible Flows readily Diffusion within a liquid occurs slowly
Solid	Retains own shape and volume Does not expand to fill its container Is virtually incompressible Does not flow Diffusion within a solid occurs extremely slowly

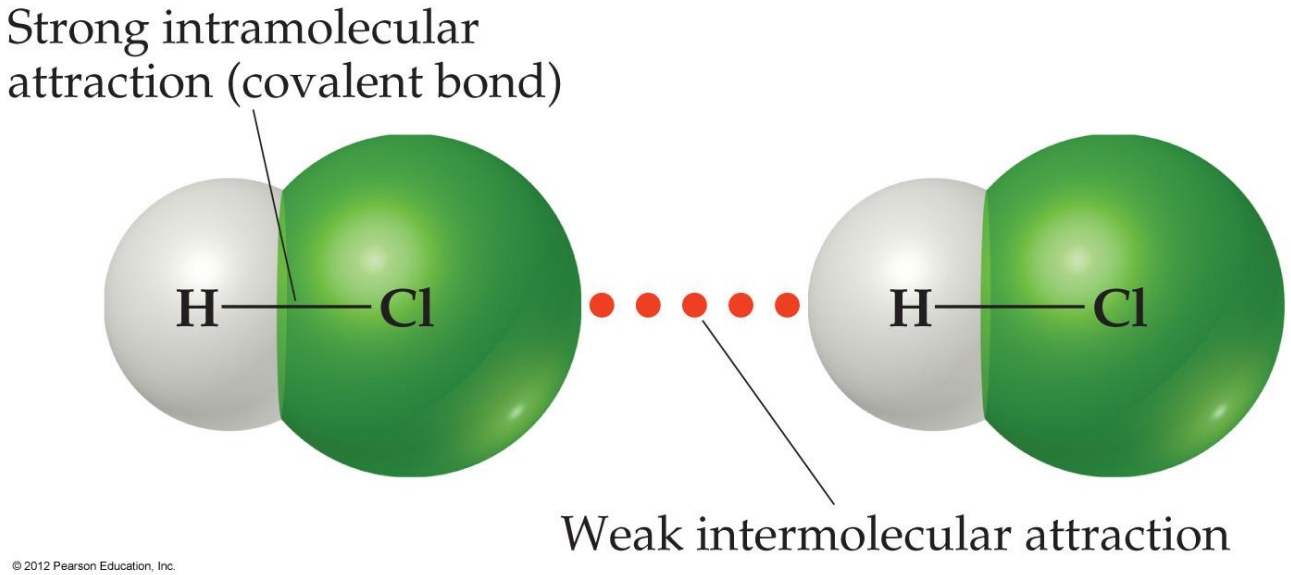
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- The state a substance is in at a particular temperature and pressure depends on two antagonistic entities:
  - The kinetic energy of the particles.
  - The strength of the attractions between the particles.



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Intermolecular Forces

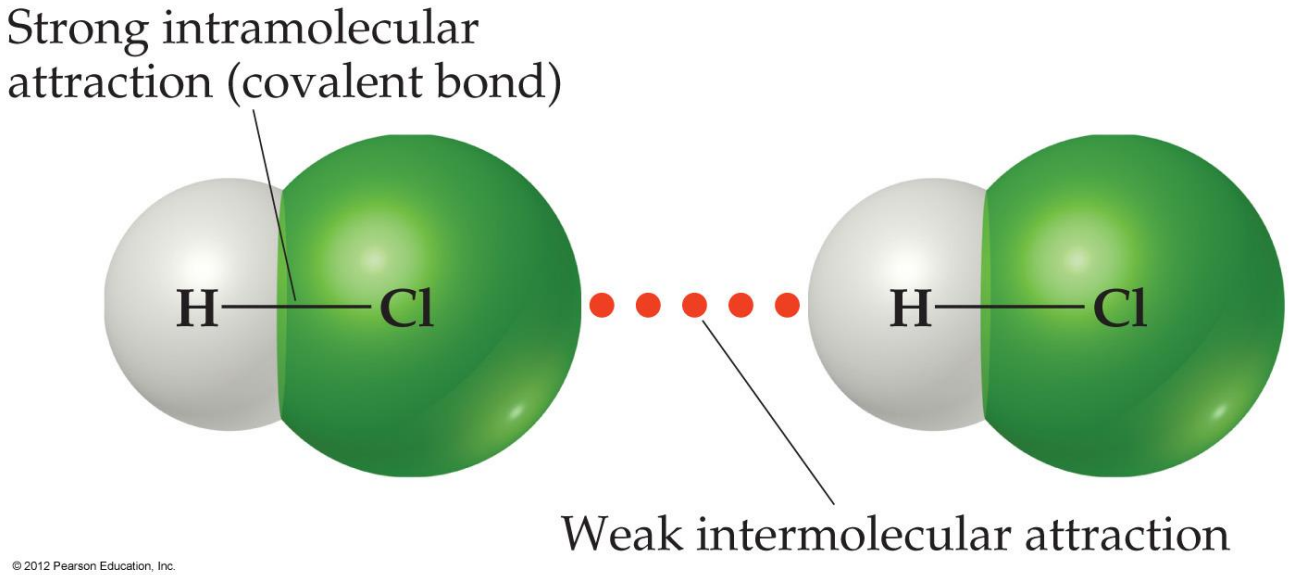


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The attractions between molecules are not nearly as strong as the intramolecular attractions that hold compounds together.

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Intermolecular Forces

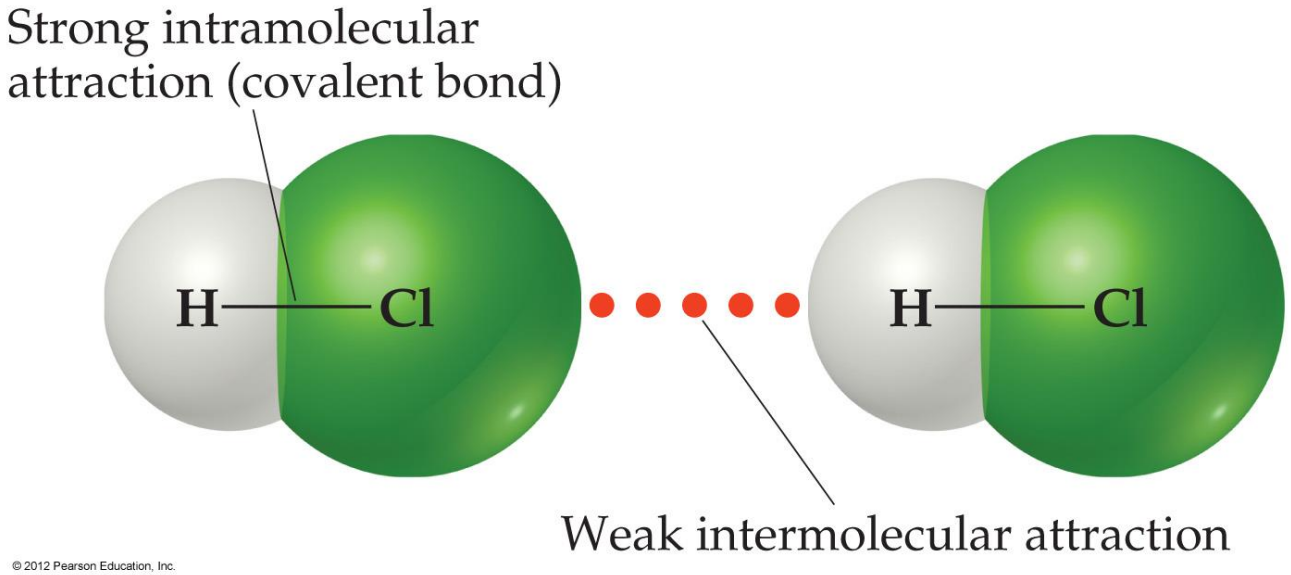


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These intermolecular attractions are, however, strong enough to control physical properties, such as boiling and melting points, vapor pressures, and viscosities.

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Intermolecular Forces



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Three types of intermolecular attractions:

London dispersion forces

Dipole-dipole attractions

Hydrogen bonding

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- London dispersion forces
- Dipole–dipole attractions

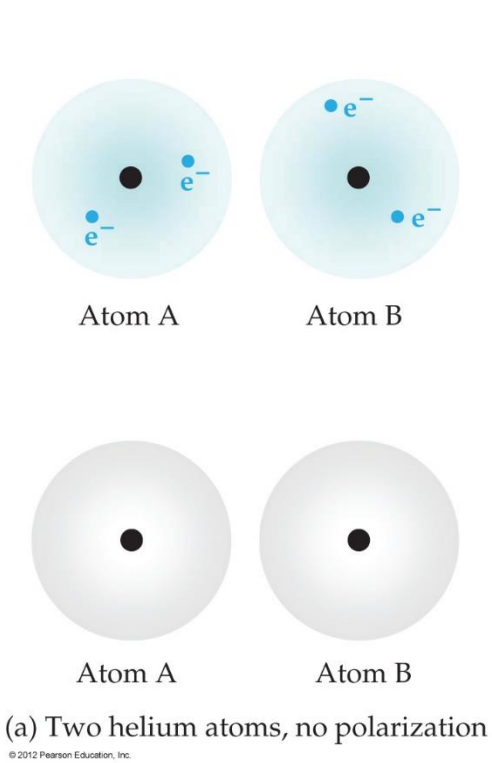
} Van der Waals forces

- Hydrogen bonding

Another kind of attractive force, the ion-dipole force, is İmportant in solutions.

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London Dispersion Forces



(a) Two helium atoms, no polarization

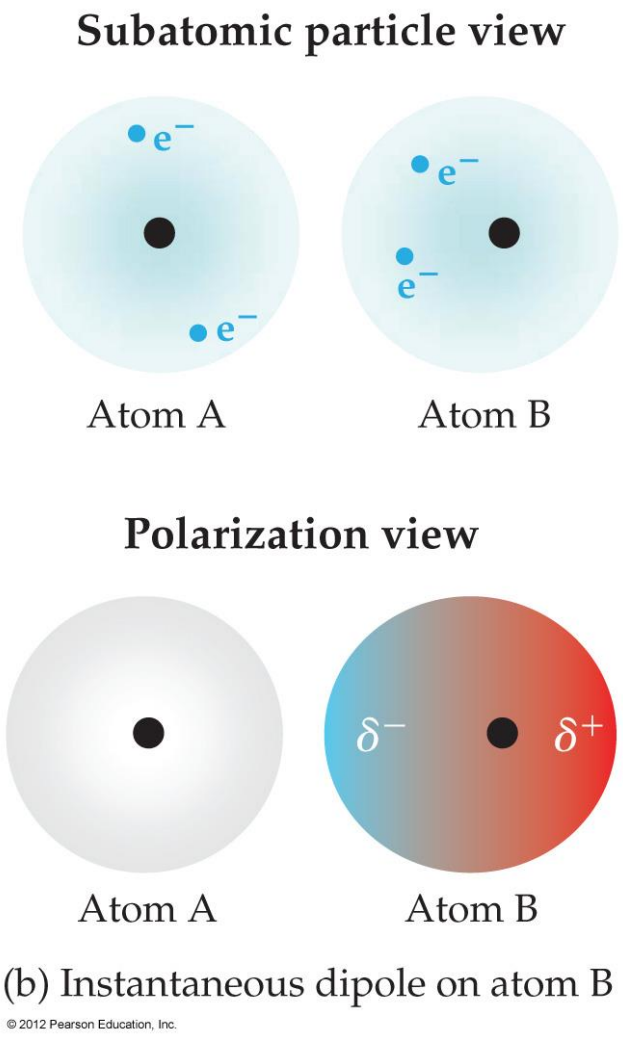
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While the electrons in the 1*s* orbital of helium would repel each other (and, therefore, tend to stay far away from each other), it does happen that they occasionally wind up on the same side of the atom.

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London Dispersion Forces

At that instant, then, the helium atom is polar, with an excess of electrons on the left side and a shortage on the right side.



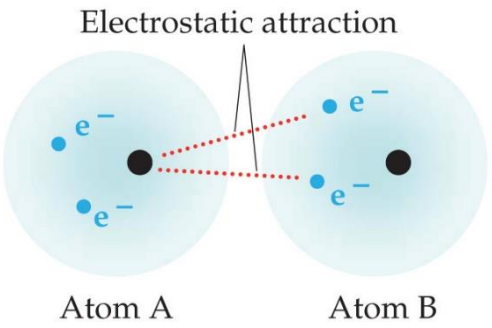
(b) Instantaneous dipole on atom B

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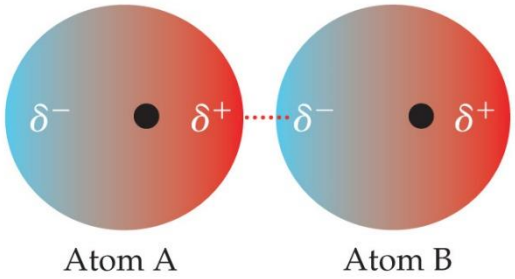
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London Dispersion Forces

Subatomic particle view



Polarization view



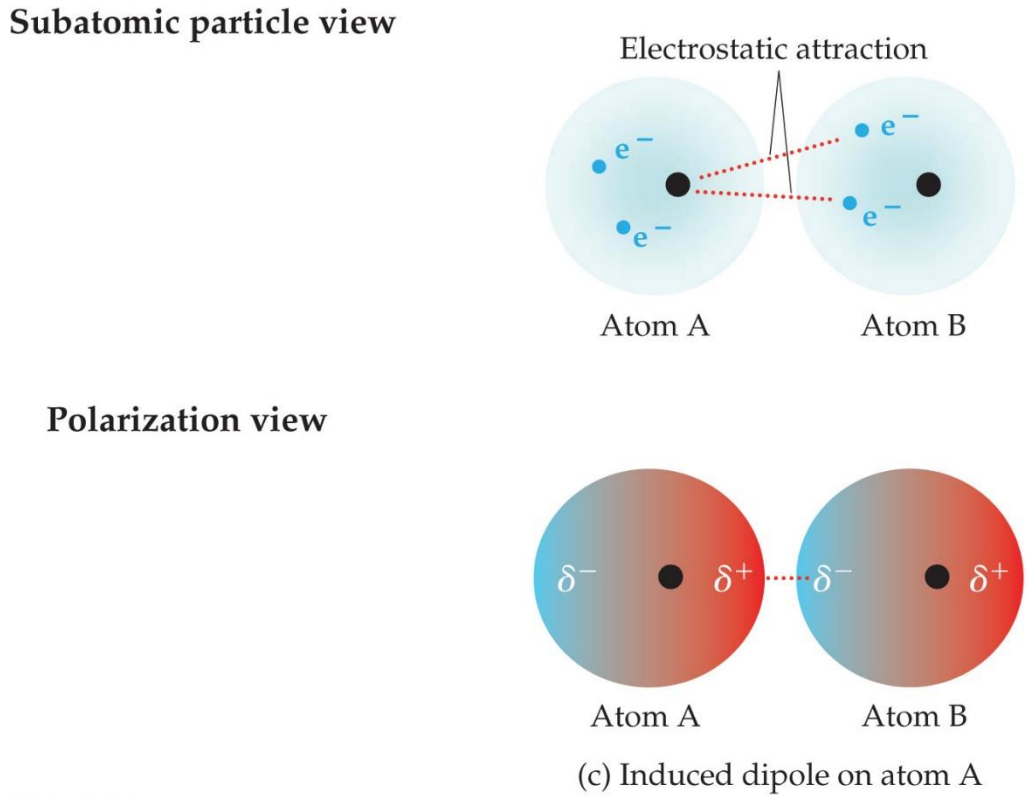
(c) Induced dipole on atom A

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Another helium atom nearby, then, would have a dipole induced in it, as the electrons on the left side of helium atom 2 repel the electrons in the cloud on helium atom 1.

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London Dispersion Forces



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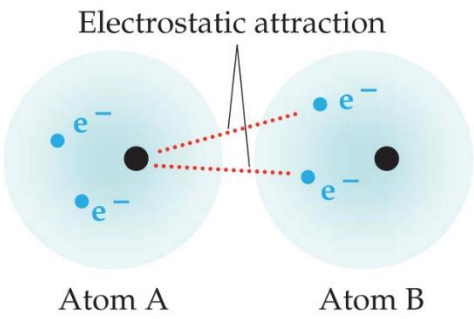
London dispersion forces, or dispersion forces, are attractions between an **instantaneous dipole and an induced dipole.**

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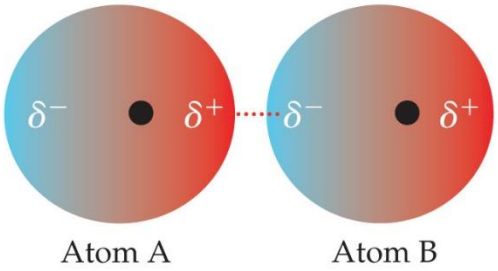
London Dispersion Forces

- These forces are present in *all* molecules, whether they are polar or nonpolar.
- The tendency of an electron cloud to distort in this way is called **polarizability**.

Subatomic particle view



Polarization view

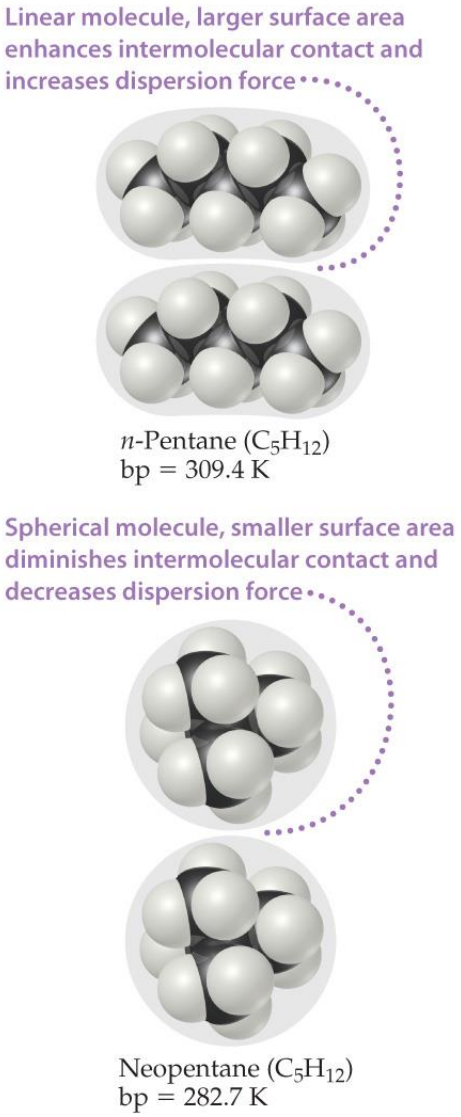


(c) Induced dipole on atom A

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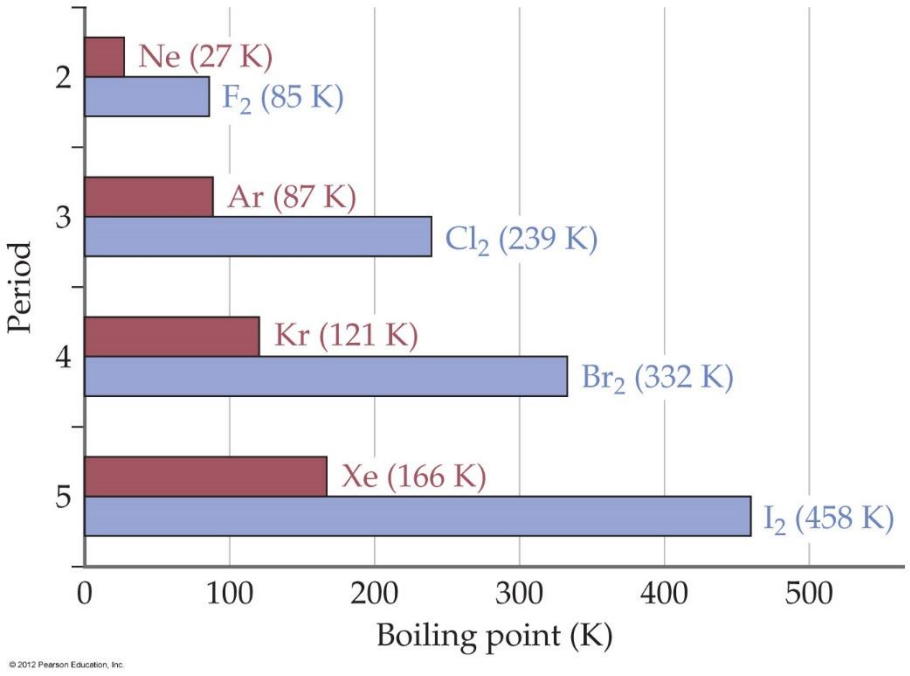
Factors Affecting London Forces



- The shape of the molecule affects the strength of dispersion forces: long, skinny molecules (like *n*-pentane) tend to have stronger dispersion forces than short, fat ones (like neopentane).
- This is due to the increased surface area in *n*-pentane.

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Factors Affecting London Forces



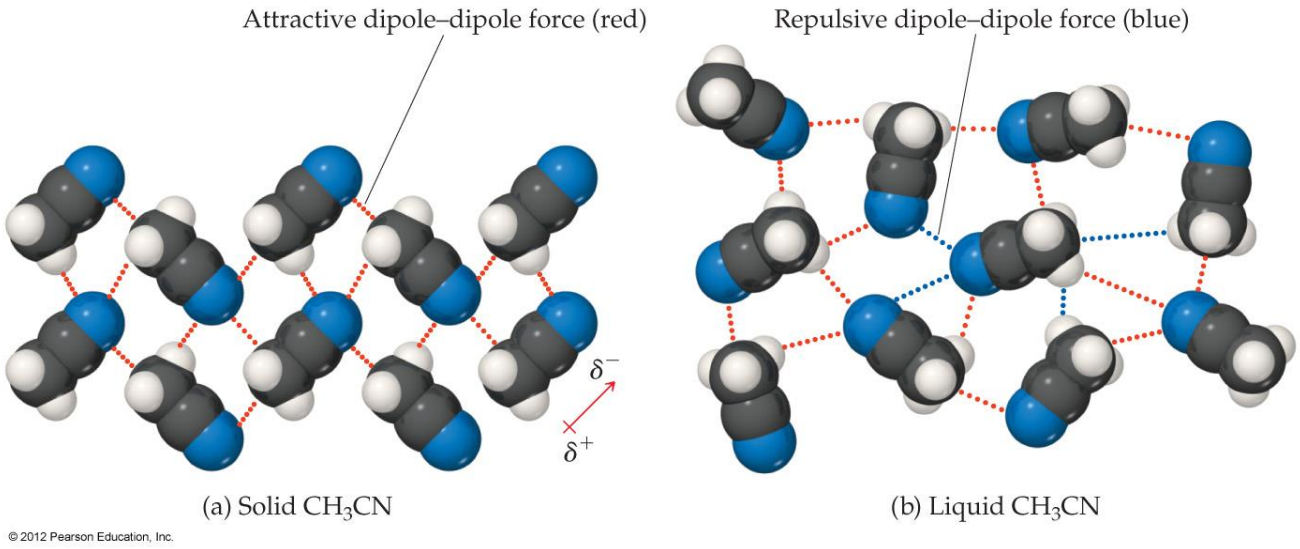
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- The strength of dispersion forces tends to increase with increased molecular weight.
- Larger atoms have larger electron clouds that are easier to polarize.

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Dipole–Dipole Interactions

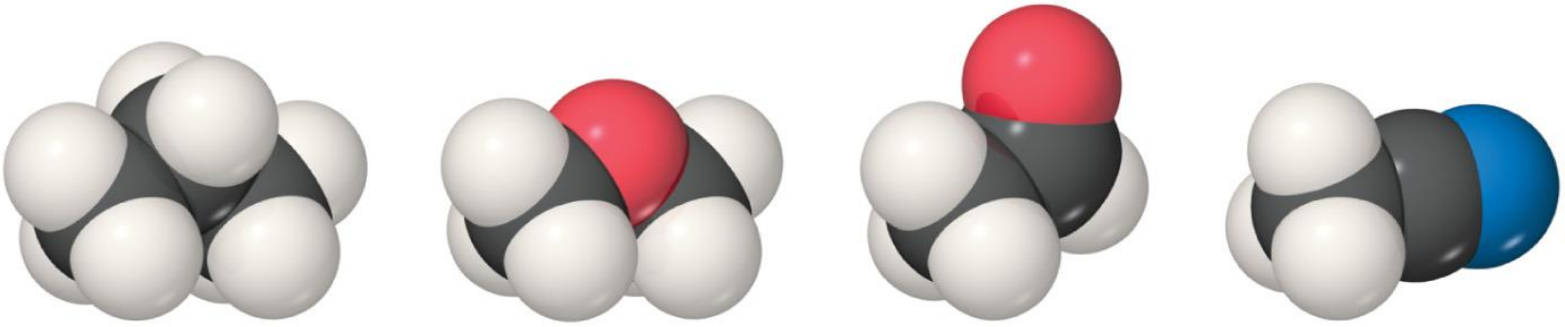
- Molecules that have permanent dipoles are attracted to each other.
  - The positive end of one is attracted to the negative end of the other, and vice versa.
  - These forces are only important when the molecules are close to each other.



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Dipole–Dipole Interactions



Propane
$CH_3CH_2CH_3$
MW = 44 amu
μ = 0.1 D
bp = 231 K

Dimethyl ether
CH_3OCH_3
MW = 46 amu
μ = 1.3 D
bp = 248 K

Acetaldehyde
CH_3CHO
MW = 44 amu
μ = 2.7 D
bp = 294 K

Acetonitrile
CH_3CN
MW = 41 amu
μ = 3.9 D
bp = 355 K

Increasing polarity
Increasing strength of dipole–dipole forces

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The more polar the molecule, the higher its boiling point.

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Which Have a Greater Effect?

Dipole–Dipole Interactions or Dispersion Forces

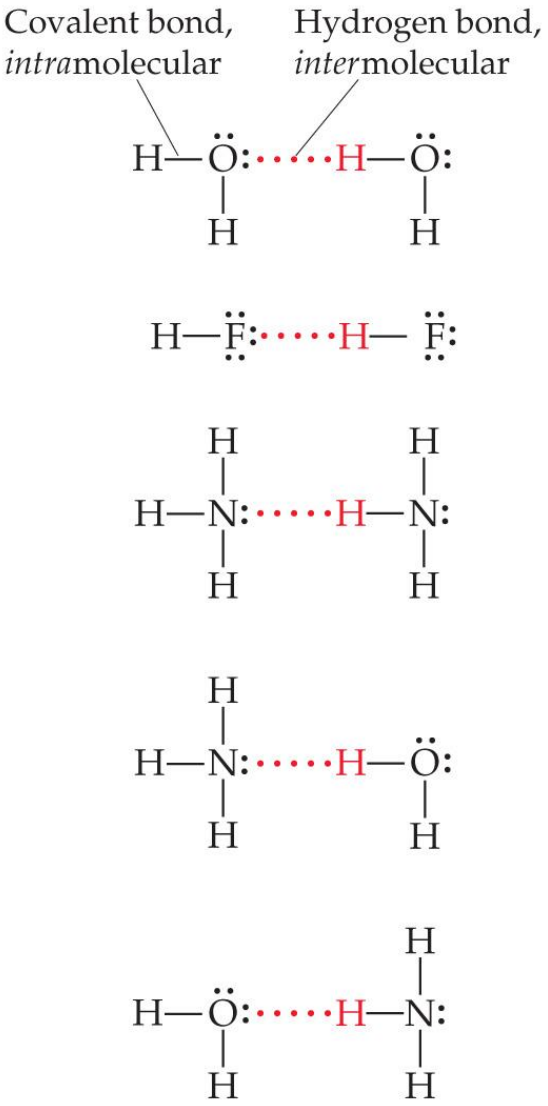
- If two molecules are of comparable size and shape, dipole–dipole interactions will likely be the dominating force.

- If one molecule is much larger than another, dispersion forces will likely determine its physical properties.

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Hydrogen Bonding

- The dipole–dipole interactions experienced when H is bonded to N, O, or F are unusually strong.
- We call these interactions **hydrogen bonds**.

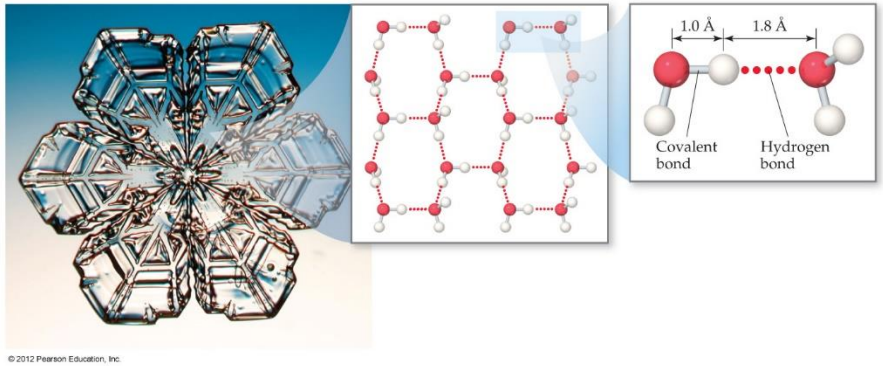


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Hydrogen Bonding

- Hydrogen bonding arises in part from the high electronegativity of nitrogen, oxygen, and fluorine.

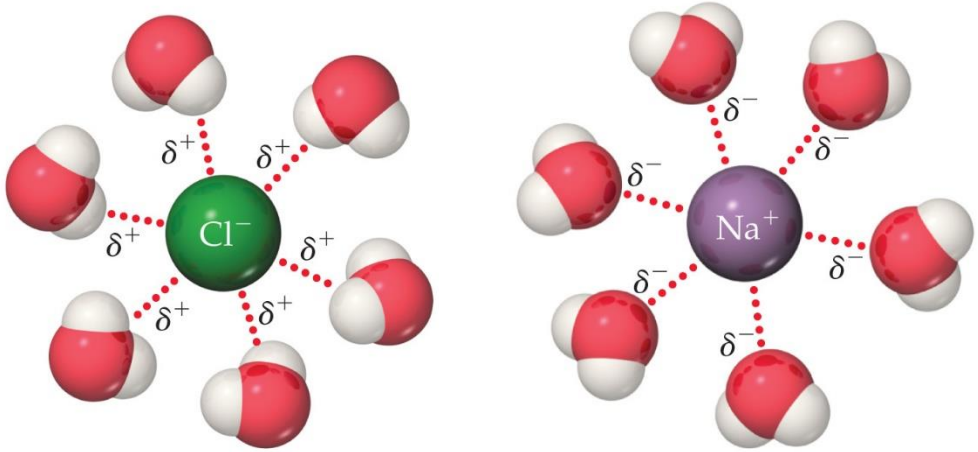


Also, when hydrogen is bonded to one of those very electronegative elements, the hydrogen nucleus is exposed.

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Ion–Dipole Interactions

- Ion–dipole interactions (a fourth type of force) are important in solutions of ions.
- The strength of these forces is what makes it possible for ionic substances to dissolve in polar solvents.



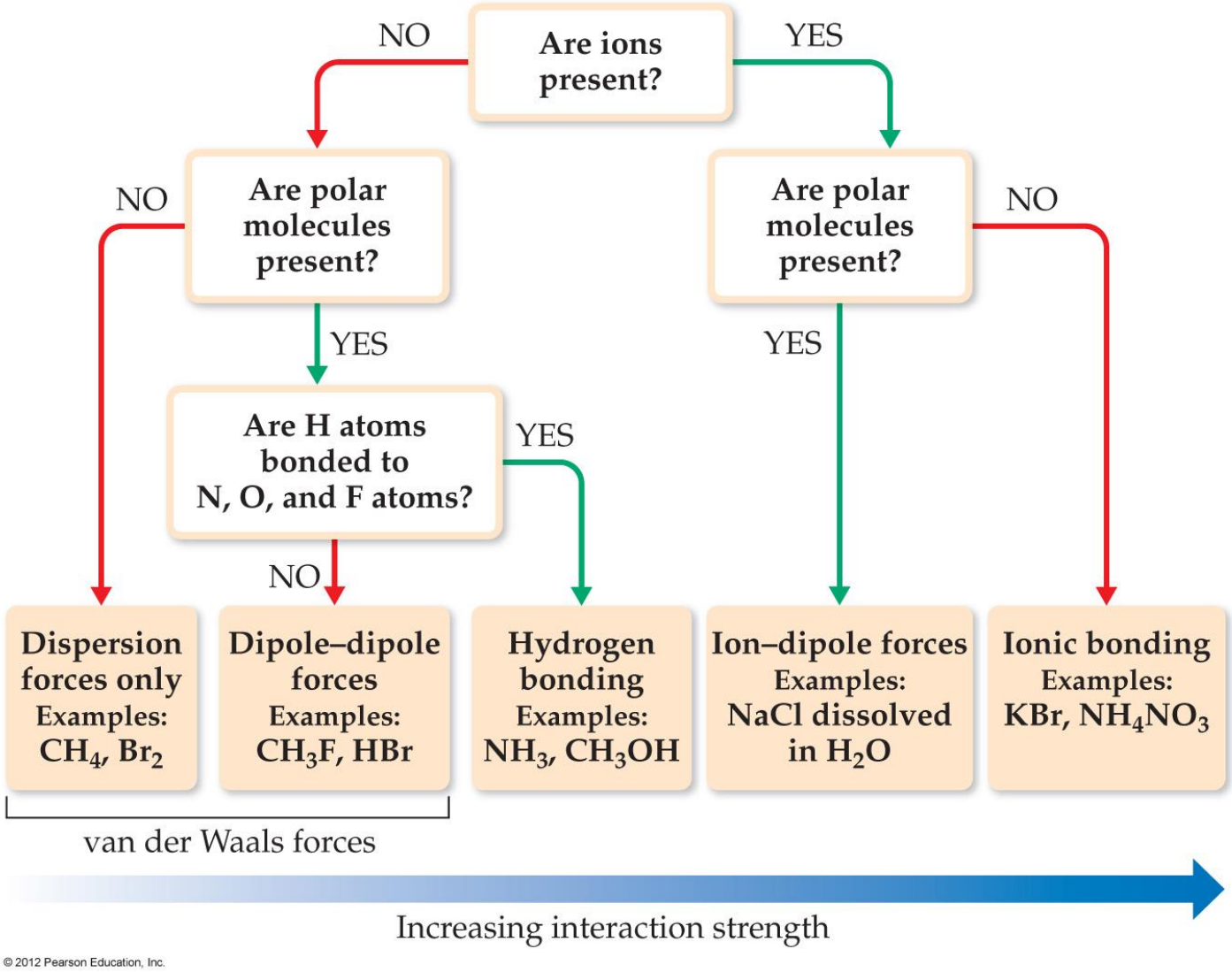
Positive ends of polar molecules are oriented toward negatively charged anion

Negative ends of polar molecules are oriented toward positively charged cation

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Summarizing Intermolecular Forces



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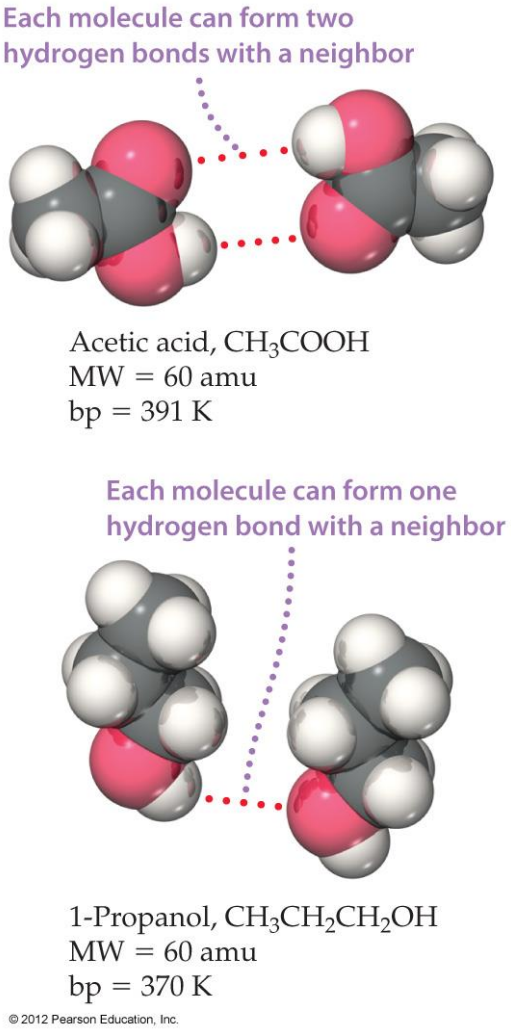
Intermolecular Forces

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Type of intermolecular interaction	Atoms Examples: Ne, Ar	Nonpolar molecules Examples: BF_3, CH_4	Polar molecules without OH, NH, or HF groups Examples: HCl, CH_3CN	Polar molecules containing OH, NH, or HF groups Examples: H_2O, NH_3	Ionic solids dissolved in polar liquids Examples: NaCl in H_2O
Dispersion forces (0.1–30 kJ/mol)	√	√	√	√	√
Dipole–dipole interactions (2–15 kJ/mol)			√	√	
Hydrogen bonding (10–40 kJ/mol)				√	
Ion–dipole interactions (>50 kJ/mol)					√

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Intermolecular Forces Affect Many Physical Properties



The strength of the attractions between particles can greatly affect the properties of a substance or solution.

Intermolecular Forces

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Viscosity

- Resistance of a liquid to flow is called **viscosity.**
- It is related to the ease with which molecules can move past each other.
- Viscosity increases with stronger intermolecular forces and decreases with higher temperature.



SAE 40
higher number
higher viscosity
slower pouring

SAE 10
lower number
lower viscosity
faster pouring

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TABLE 11.4 • Viscosities of a Series of Hydrocarbons at 20 °C

Substance	Formula	Viscosity (kg/m-s)
Hexane	$CH_3CH_2CH_2CH_2CH_2CH_3$	3.26×10^{-4}
Heptane	$CH_3CH_2CH_2CH_2CH_2CH_2CH_3$	4.09×10^{-4}
Octane	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_3$	5.42×10^{-4}
Nonane	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$	7.11×10^{-4}
Decane	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$	1.42×10^{-3}

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The viscosity of a substance decreases with increasing temperature. This can be seen in the viscosity of octane:

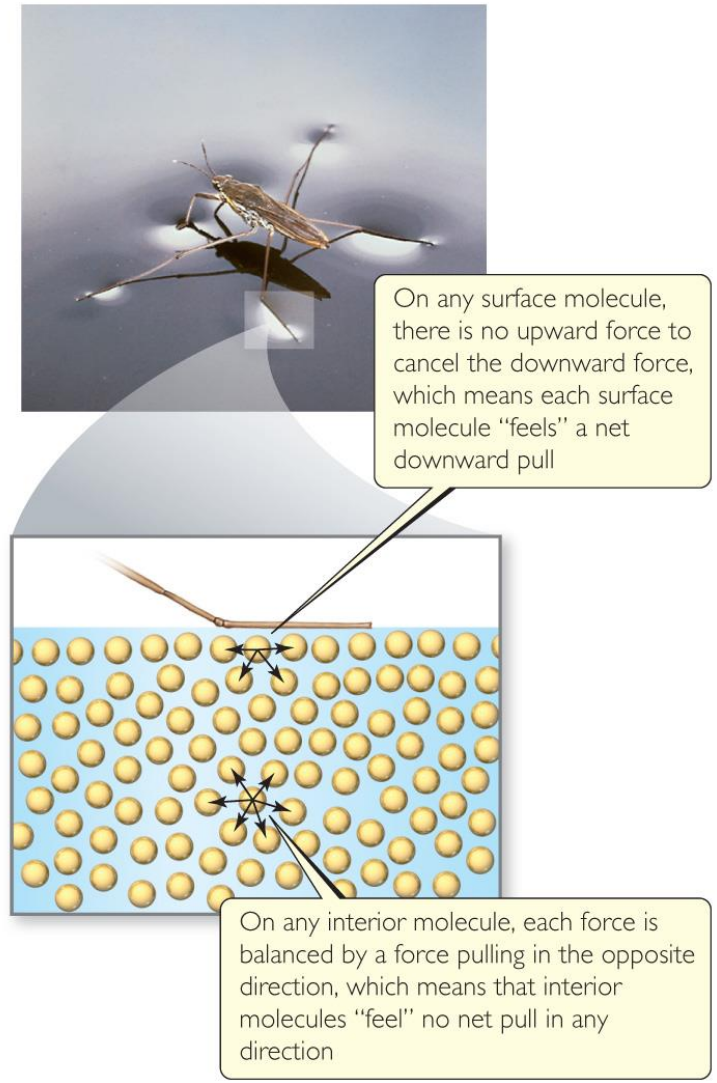
$$7.06 \times 10^{-4}\,\text{kg/m-s at } 0\,^{\circ}\text{C}$$

$$4.33 \times 10^{-4}\,\text{kg/m-s at } 40\,^{\circ}\text{C}$$

At higher temperatures, the greater average kinetic energy of the molecules overcomes the attractive forces between molecules.

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Surface Tension



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Surface tension results from the net inward force experienced by the molecules on the surface of a liquid.

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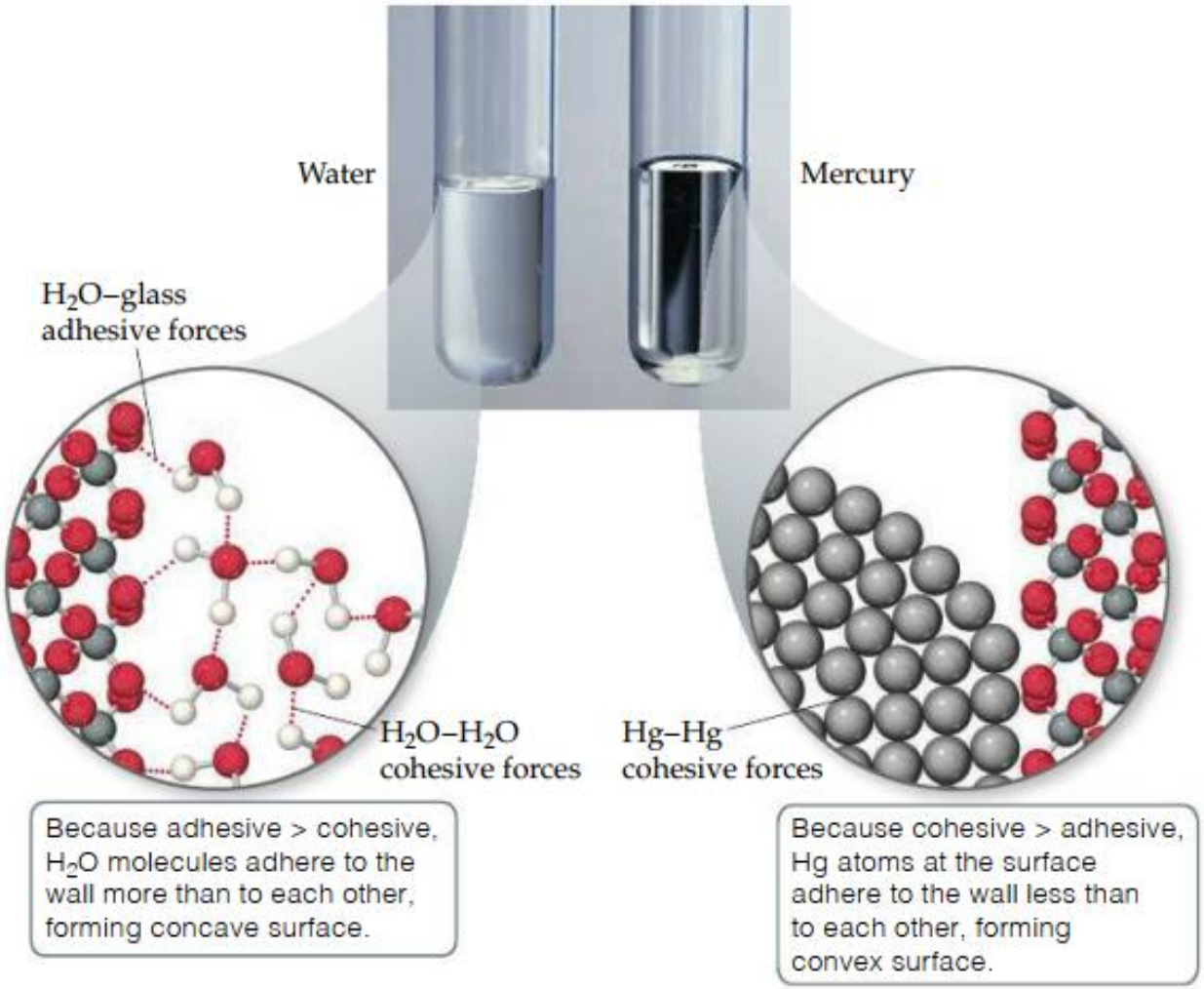
A measure of the net inward force that must be overcome to expand the surface area of a liquid is given by its surface tension. **Surface tension** is the energy required to increase the surface area of a liquid by a unit amount. For example, the surface tension of water at 20 °C is 7.29×10^{-2} J/m^2, which means that an energy of 7.29×10^{-2} J must be supplied to increase the surface area of a given amount of water by 1 m^2. Water has a high surface tension because of its strong hydrogen bonds. The surface tension of mercury is even higher (4.6×10^{-1} J/m^2) because of even stronger metallic bonds between the atoms of mercury.

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Capillary Action

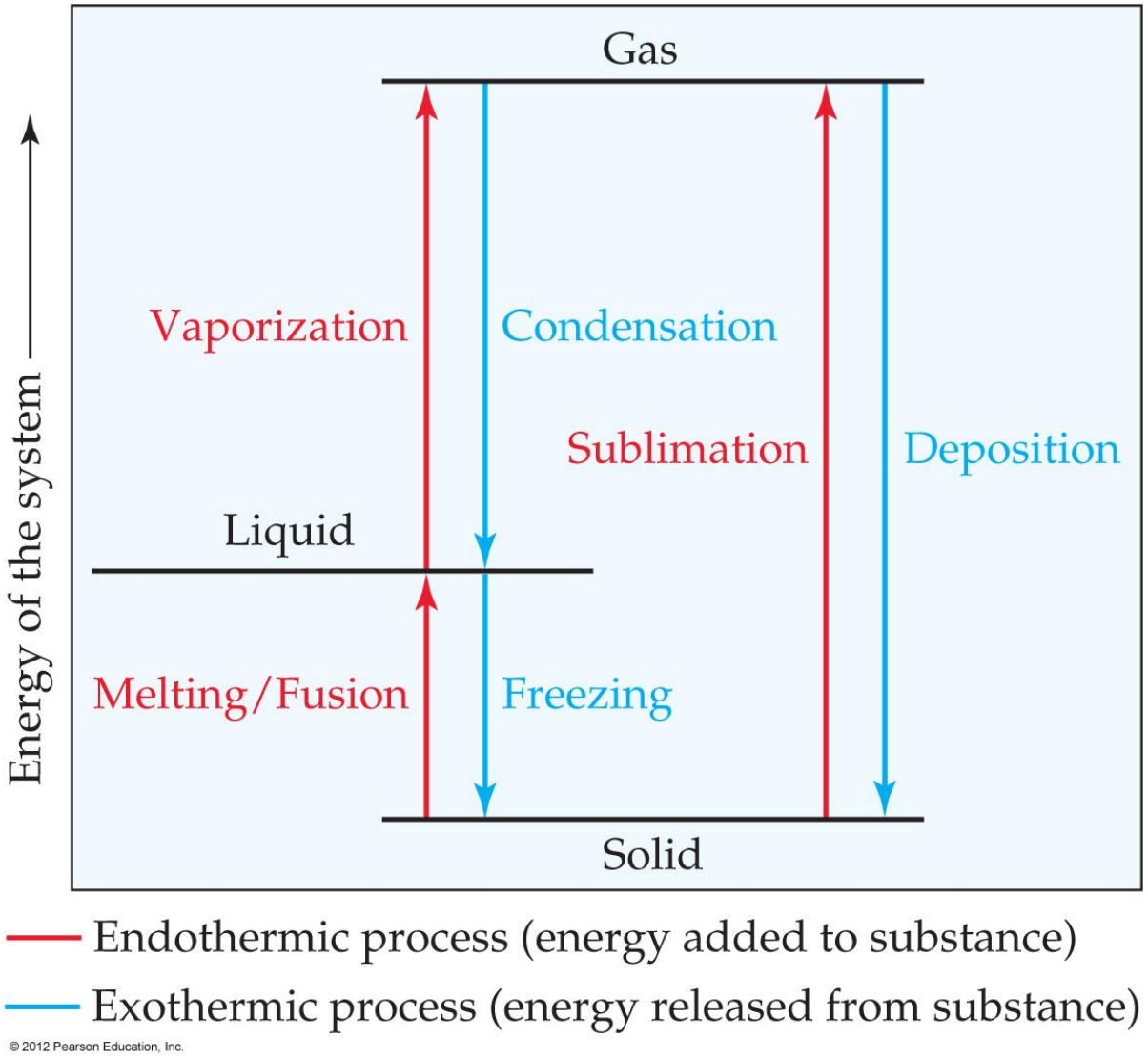
Intermolecular forces that bind similar molecules to one another, such as the hydrogen bonding in water, are called *cohesive forces*. Intermolecular forces that bind a substance to a surface are called *adhesive forces*. Water placed in a glass tube adheres to the glass because the adhesive forces between the water and the glass are greater than the cohesive forces between water molecules; glass is principally SiO_2, which has a very polar surface. The curved surface, or *meniscus*, of the water is therefore U-shaped (Figure 11.19). For mercury, however, the situation is different. Mercury atoms can form bonds with one another but not with the glass. As a result, the cohesive forces are much greater than the adhesive forces and the meniscus is shaped like an inverted U.

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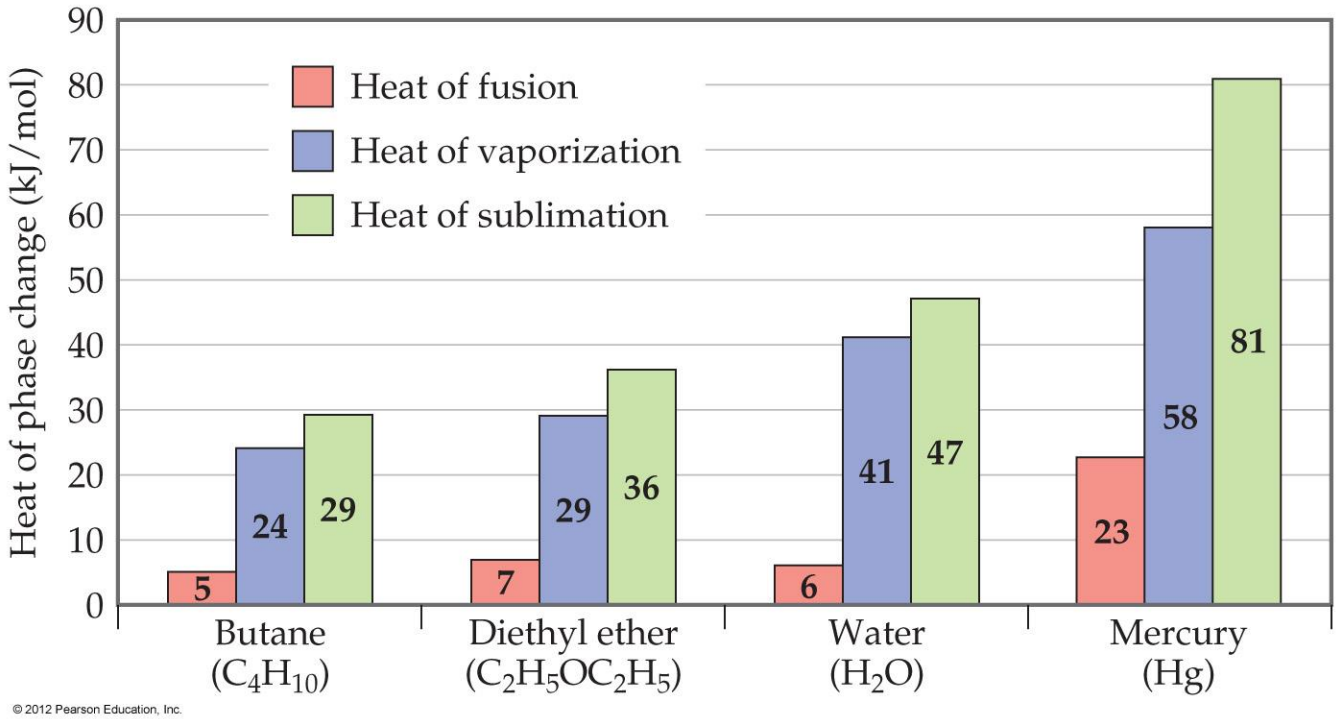
Phase Changes



Intermolecular Forces

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Energy Changes Associated with Changes of State



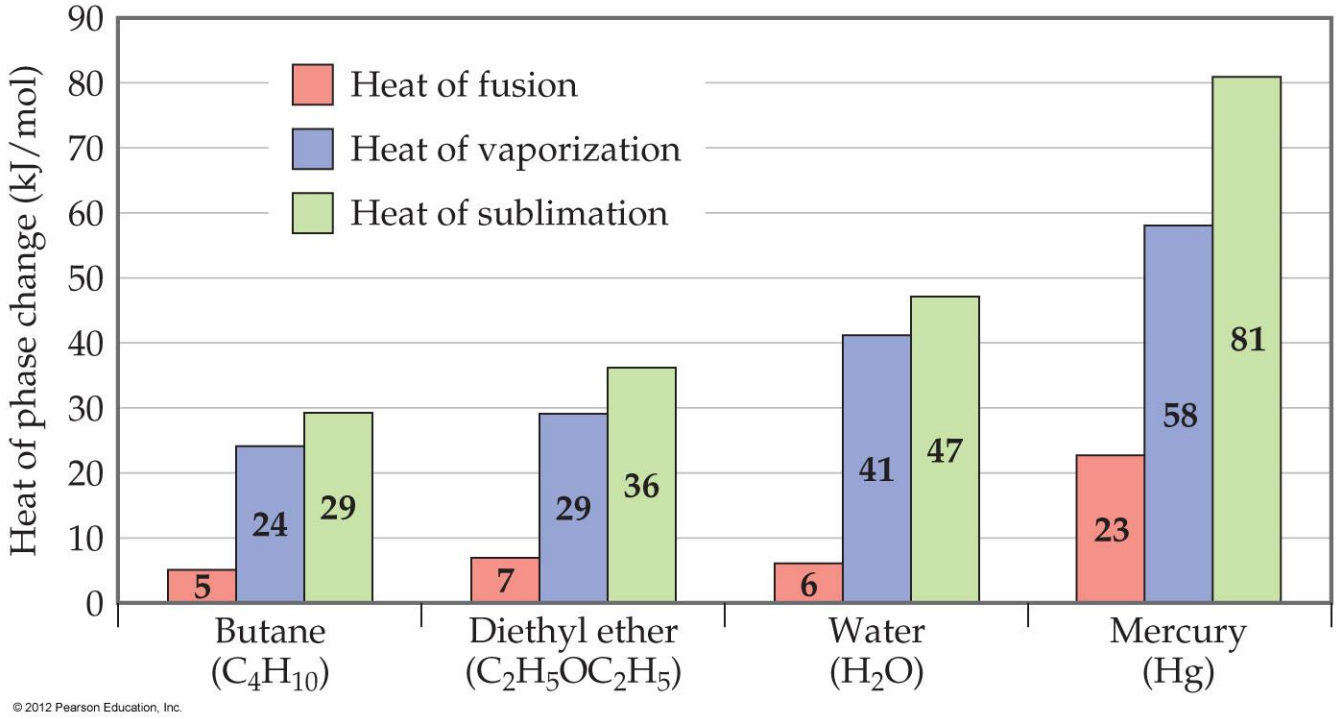
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The **heat of fusion** is the energy required to change a solid at its melting point to a liquid.

Intermolecular Forces

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Energy Changes Associated with Changes of State

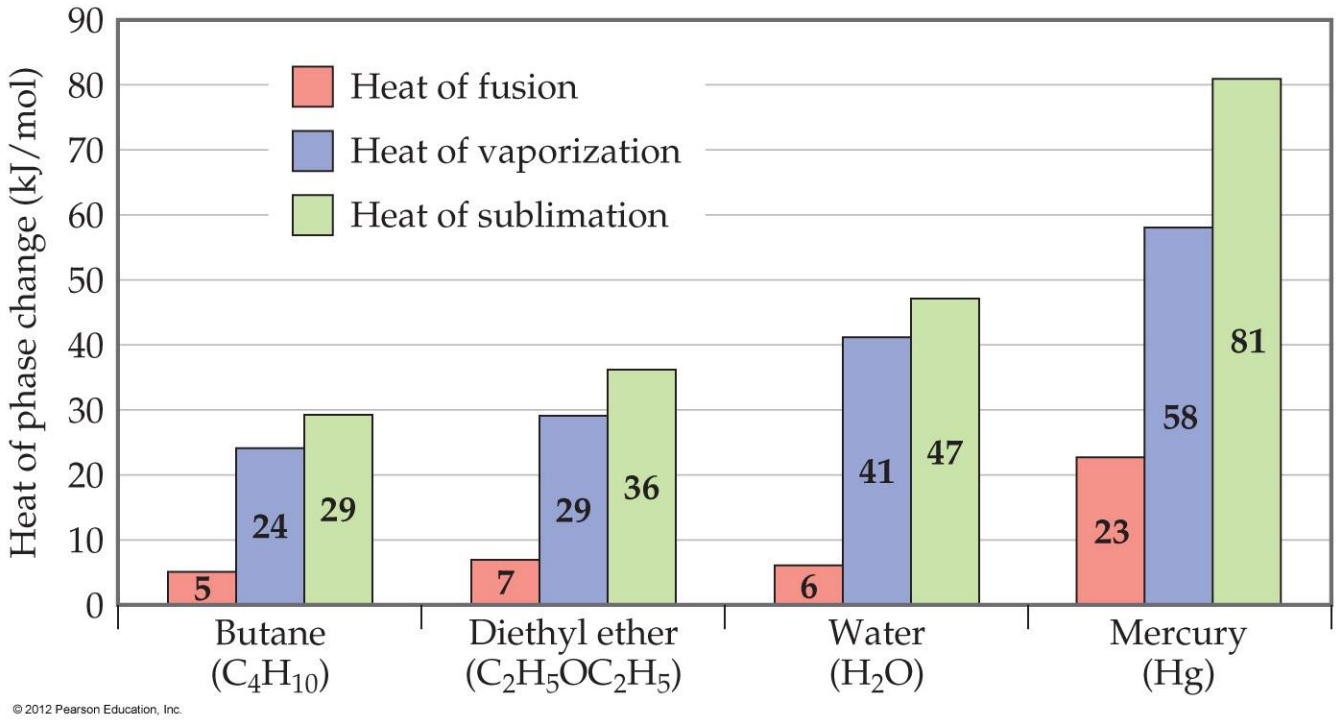


The **heat of vaporization** is defined as the energy required to change a liquid at its boiling point to a gas.

Intermolecular Forces

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Energy Changes Associated with Changes of State



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The **heat of sublimation** is defined as the energy required to change a solid directly to a gas.

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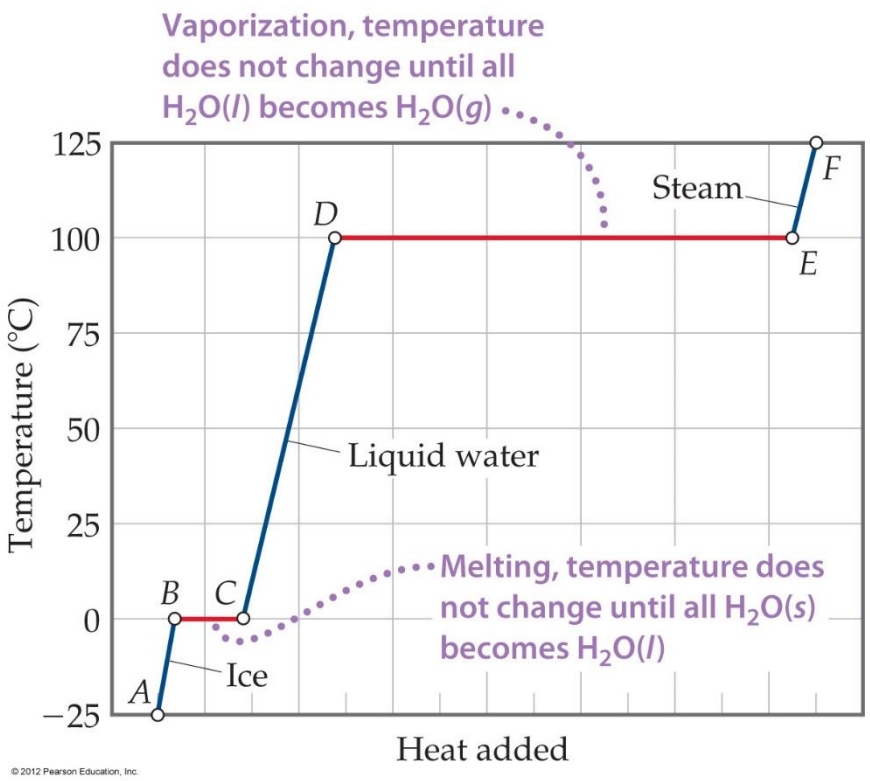
$$H_2O(s) \rightarrow H_2O(l) \qquad \Delta H_{fus} = 6.01 \text{ kJ}$$

$$H_2O(l) \longrightarrow H_2O(g) \qquad \Delta H_{vap} = 40.7 \text{ kJ}$$

The particles of a solid can move directly into the gaseous state. The enthalpy change required for this transition is called the **heat of sublimation**, denoted ΔH_{sub}. As illustrated in Figure 11.20, ΔH_{sub} is the sum of ΔH_{fus} and ΔH_{vap}. Thus, ΔH_{sub} for water is approximately 47 kJ/mol.

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Energy Changes Associated with Changes of State



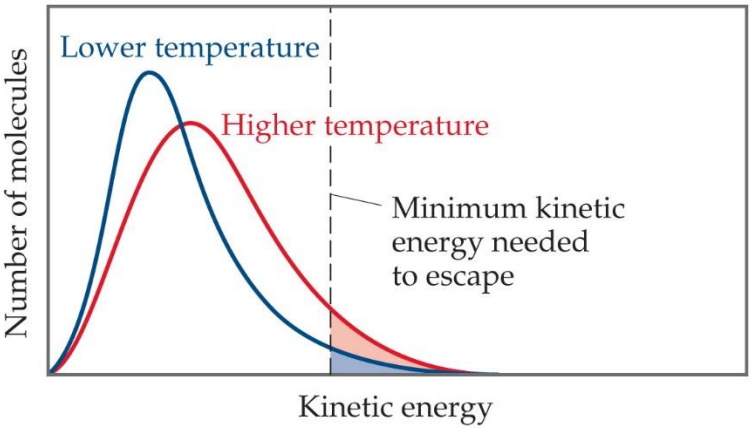
- The heat added to the system at the melting and boiling points goes into pulling the molecules farther apart from each other.
- The temperature of the substance does not rise during a phase change.

Intermolecular Forces

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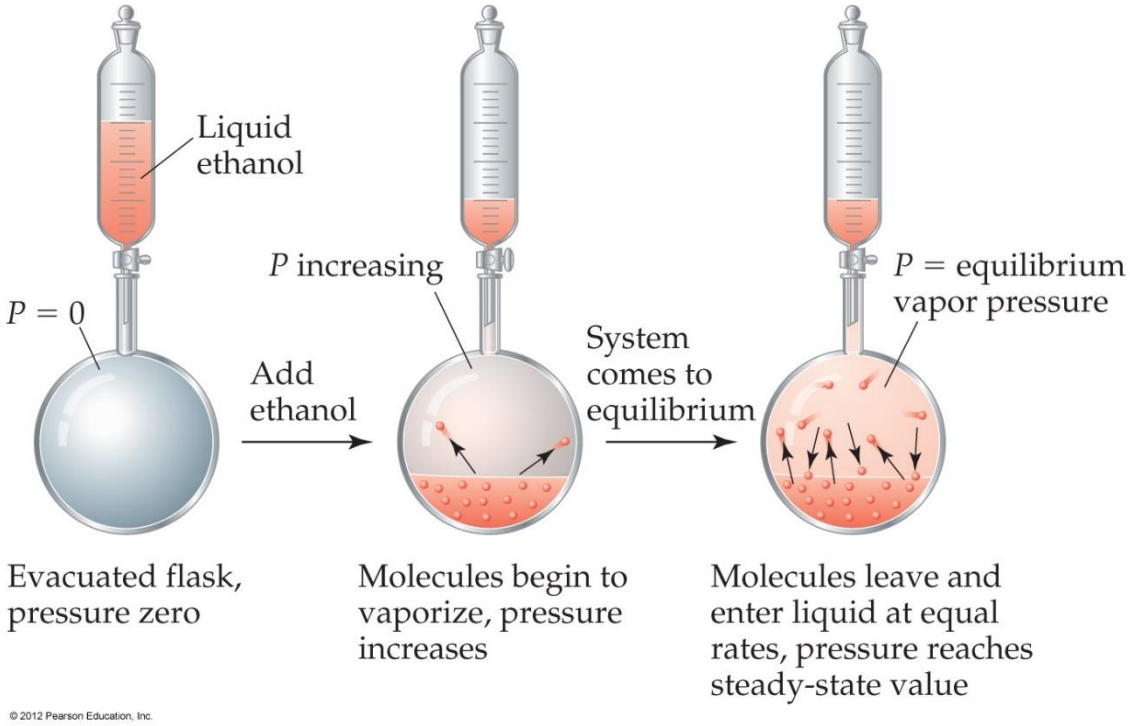
Vapor Pressure

- At any temperature some molecules in a liquid have enough energy to break free.
- As the temperature rises, the fraction of molecules that have enough energy to break free increases.



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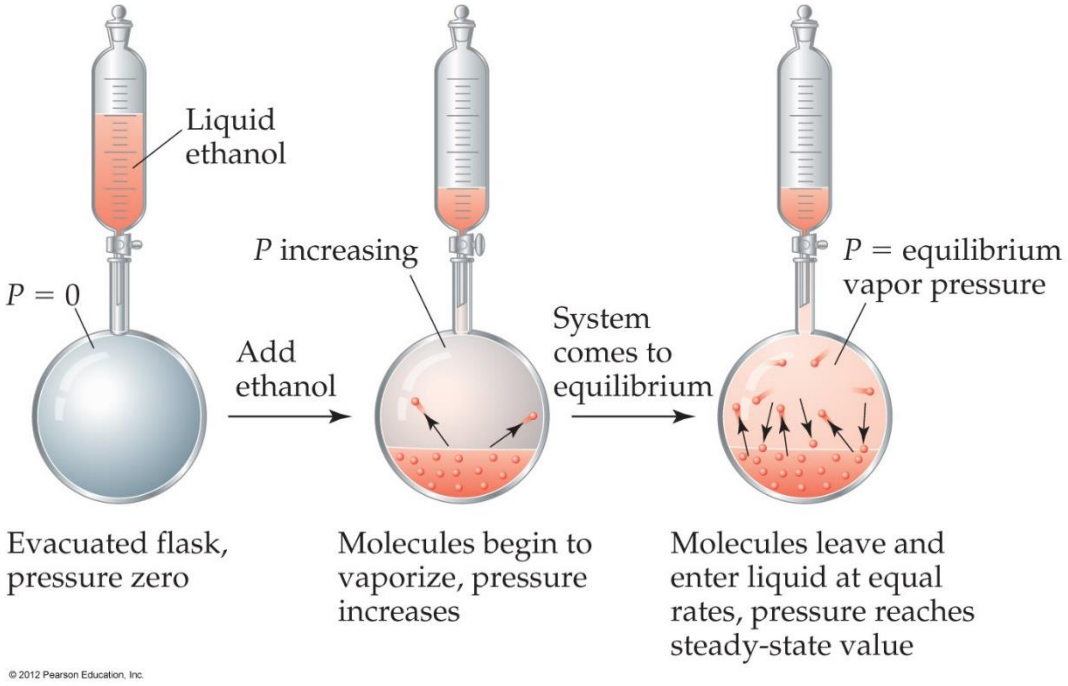
Vapor Pressure



As more molecules escape the liquid, the pressure they exert increases.

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Vapor Pressure



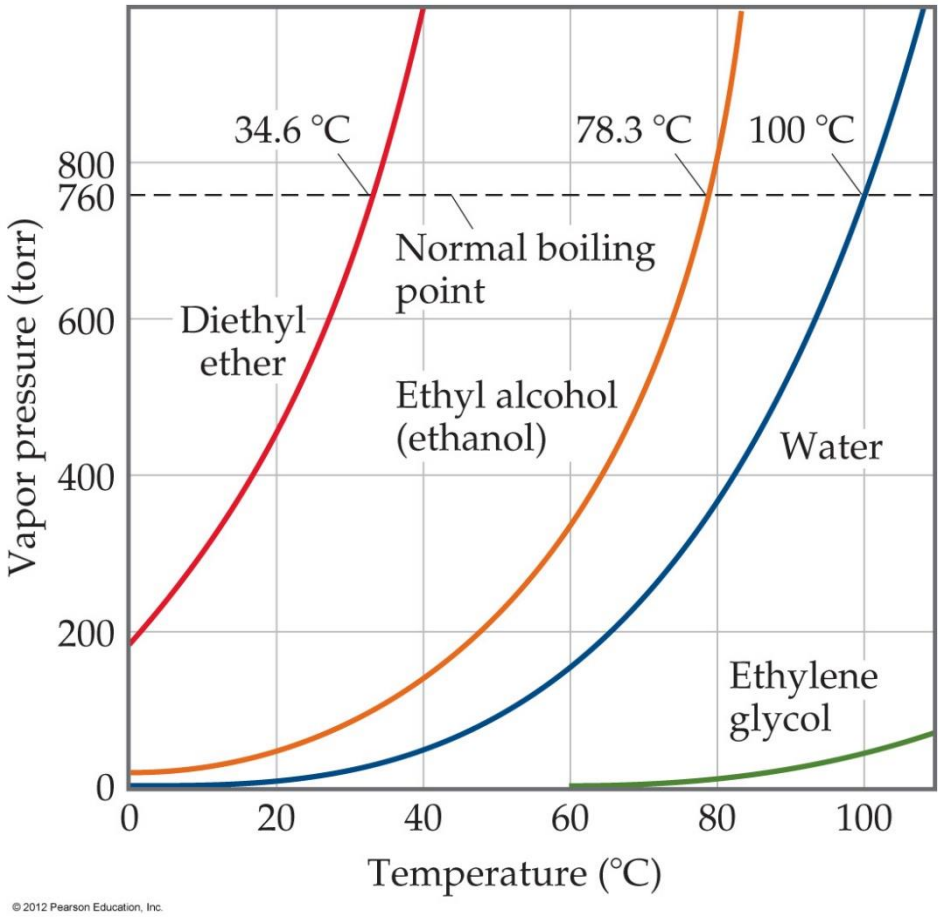
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The liquid and vapor reach a state of dynamic equilibrium: liquid molecules evaporate and vapor molecules condense *at the same rate.*

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Vapor Pressure

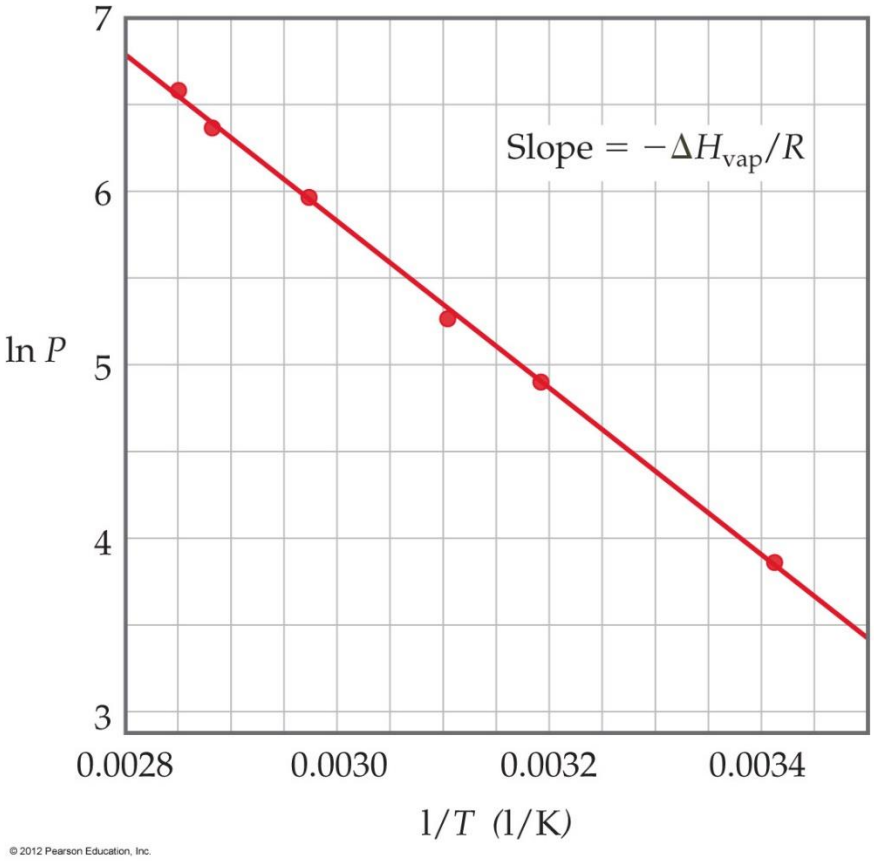
- The **boiling point** of a liquid is the temperature at which its vapor pressure equals atmospheric pressure.
- The **normal boiling point** is the temperature at which its vapor pressure is 760 torr.



Intermolecular Forces

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Vapor Pressure

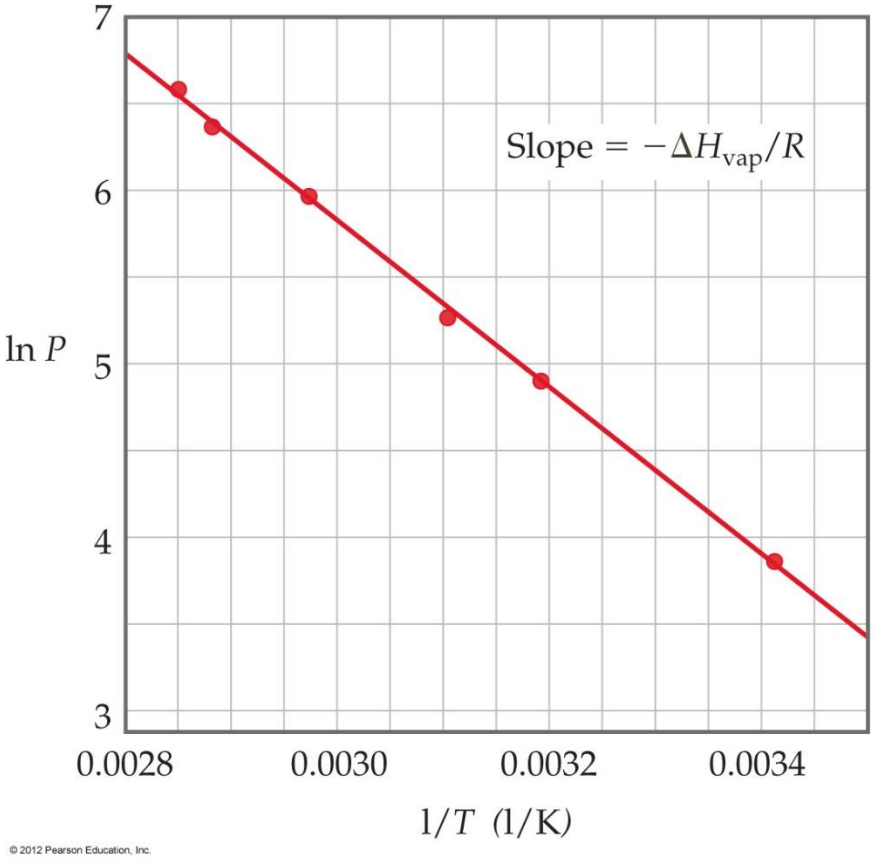


The natural log of the vapor pressure of a liquid is inversely proportional to its temperature.

Intermolecular Forces

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Vapor Pressure



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This relationship is quantified in the **Clausius–Clapeyron equation:**

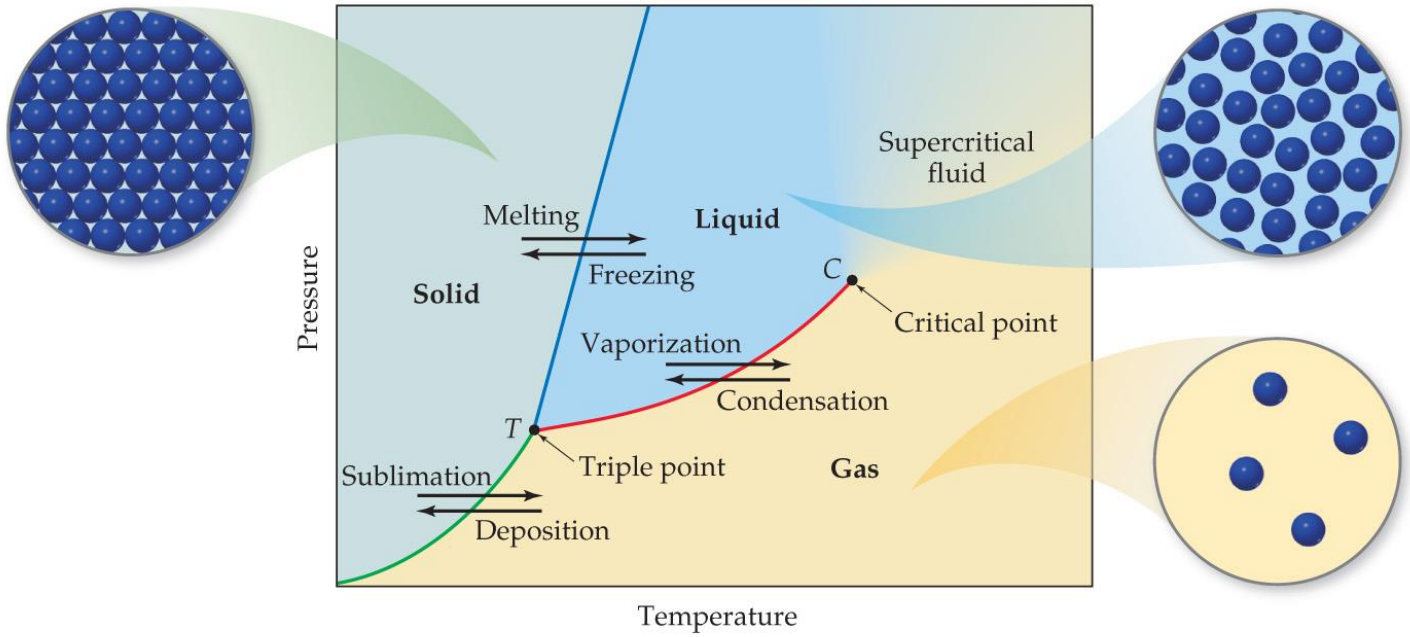
$$\ln P = -\Delta H_{vap}/RT + C,$$

where C is a constant.

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Phase Diagrams

Phase diagrams display the state of a substance at various pressures and temperatures, and the places where equilibria exist between phases.

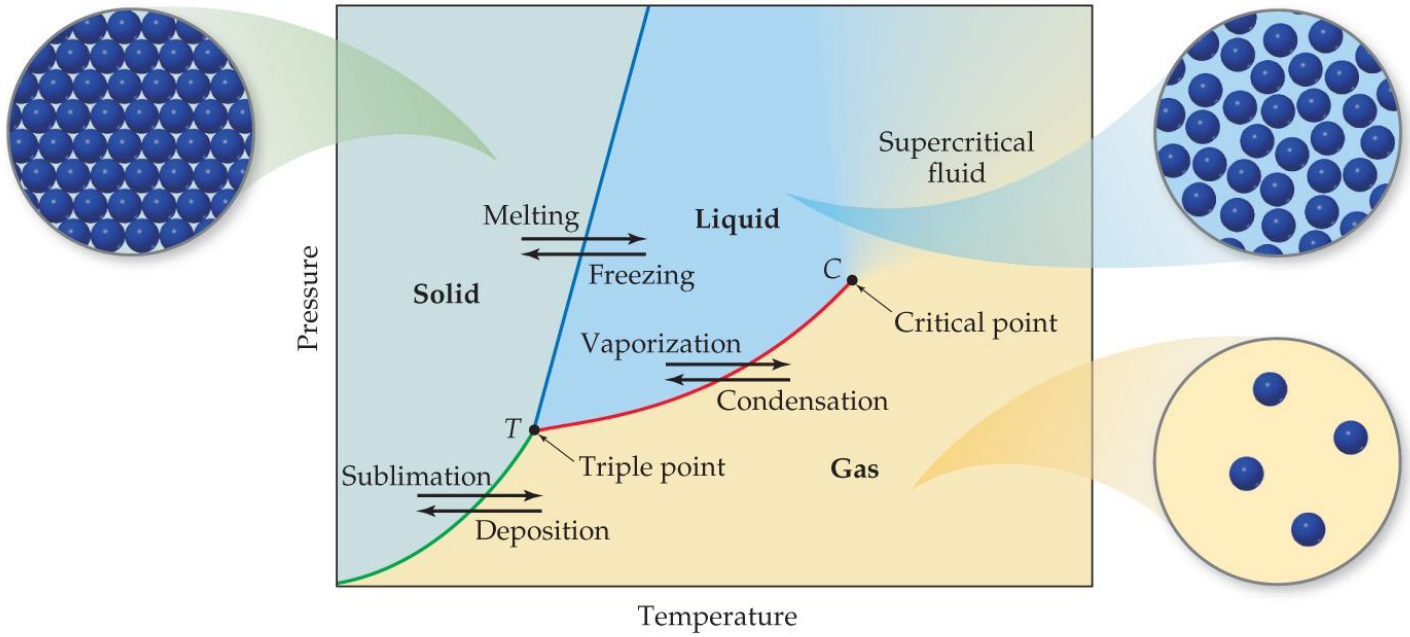


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Phase Diagrams

The liquid–vapor interface starts at the triple point (*T*), at which all three states are in equilibrium, and ends at the critical point (*C*), above which the liquid and vapor are indistinguishable from each other.



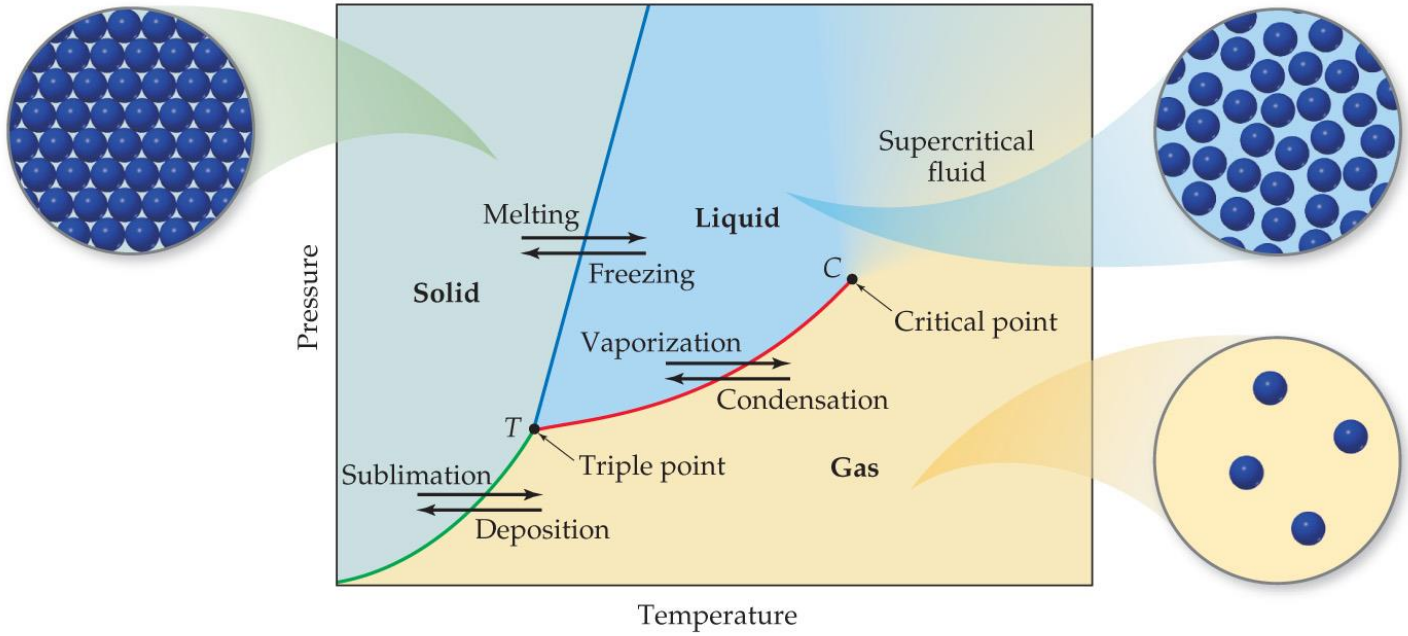
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Phase Diagrams

Each point along this line is the boiling point of the substance at that pressure.



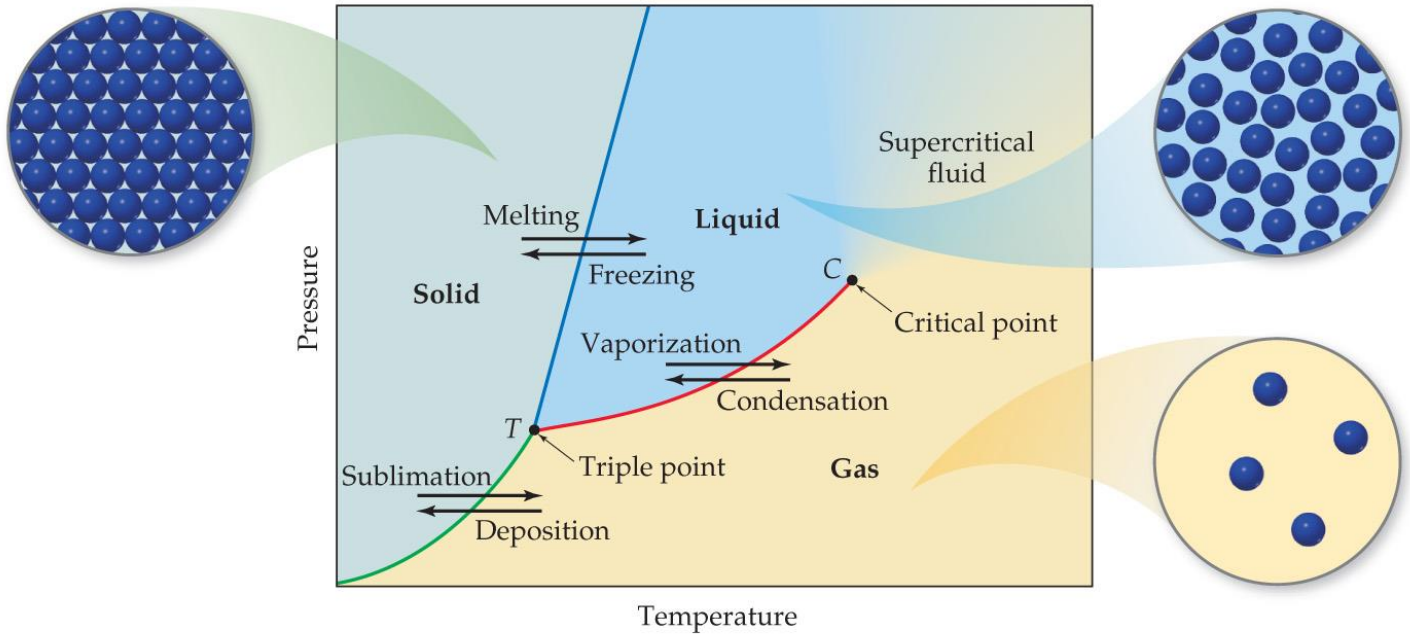
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Phase Diagrams

The interface between liquid and solid marks the melting point of a substance at each pressure.



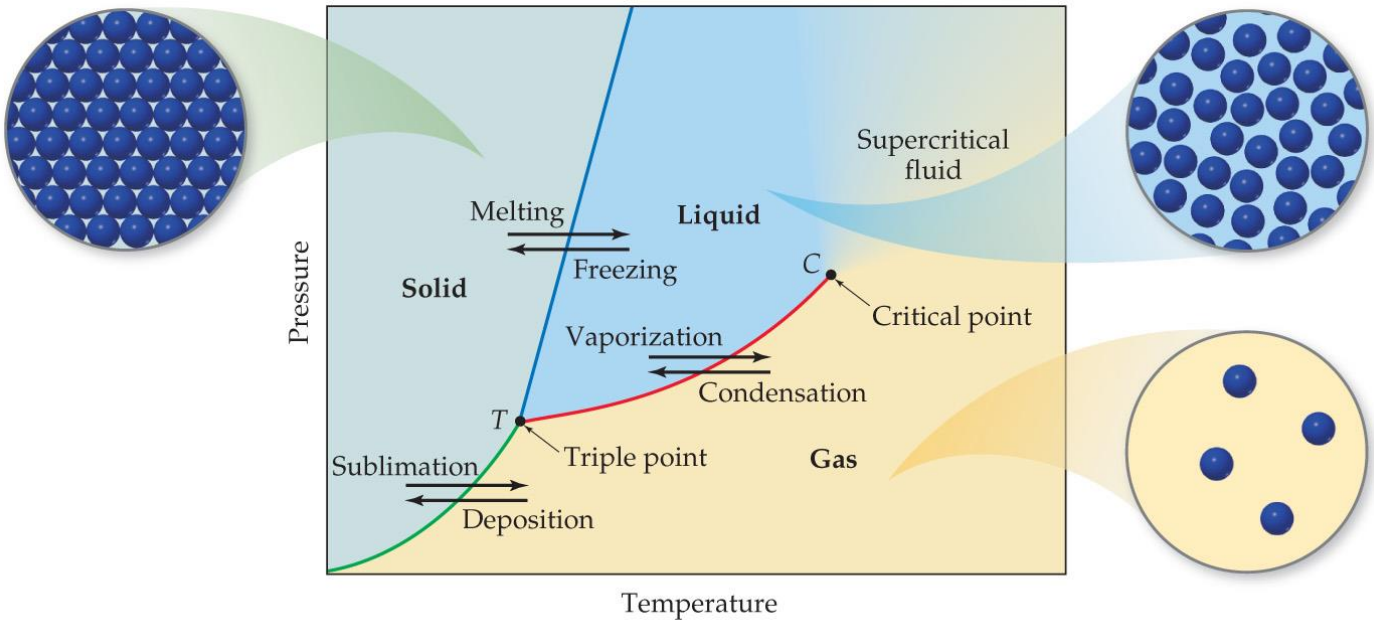
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Phase Diagrams

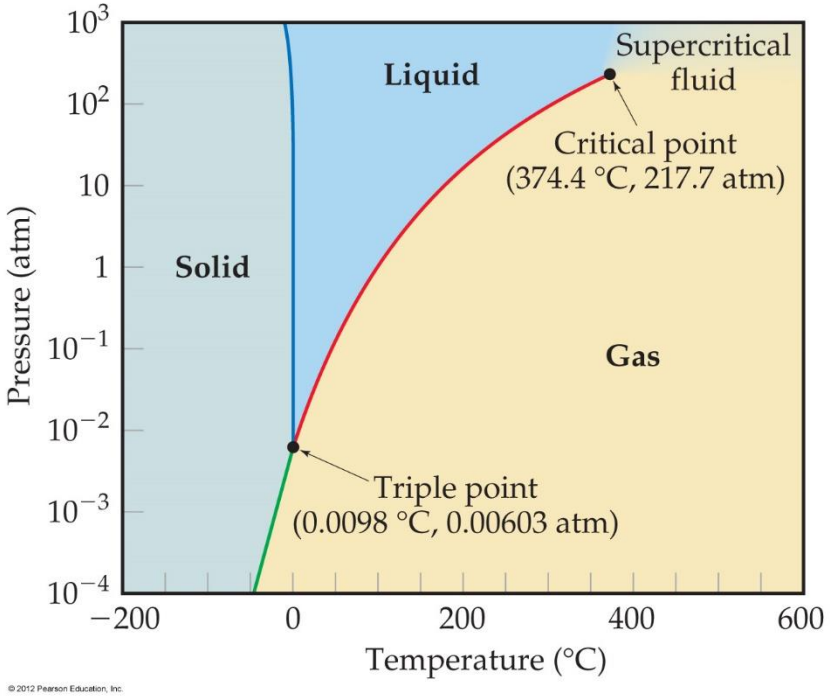
- Below the triple point the substance cannot exist in the liquid state.
- Along the solid–gas line those two phases are in equilibrium; the sublimation point at each pressure is along this line.



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Phase Diagram of Water

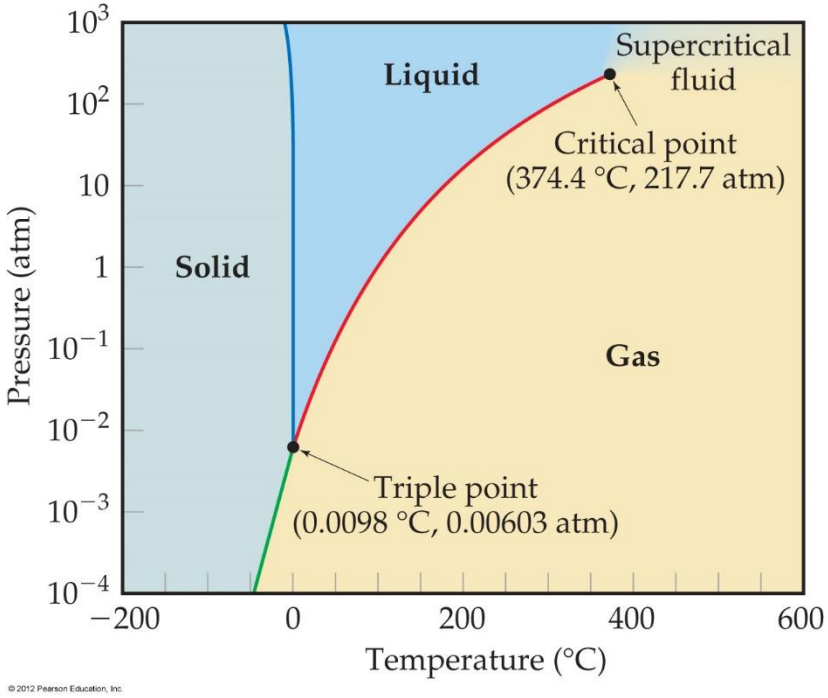


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- Note the high critical temperature and critical pressure.
  - These are due to the strong van der Waals forces between water molecules.

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Phase Diagram of Water



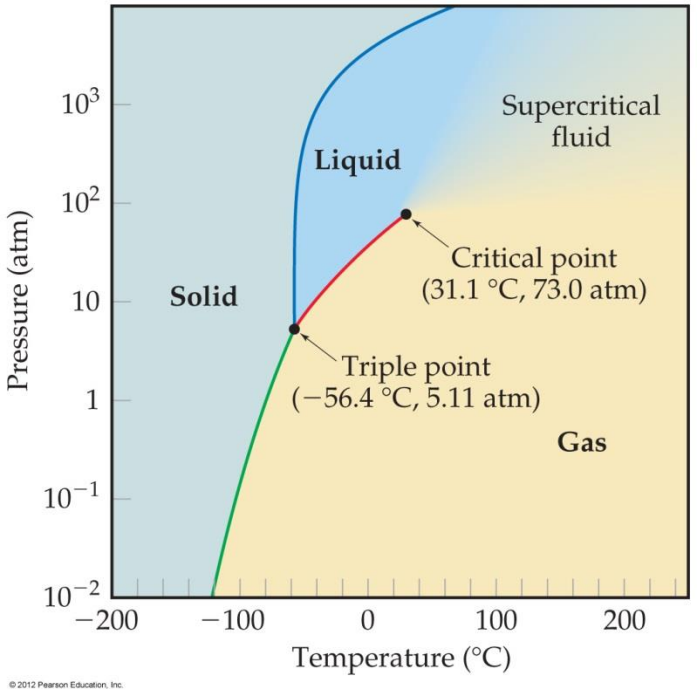
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- The slope of the solid–liquid line is negative.
  - This means that as the pressure is increased at a temperature just below the melting point, water goes from a solid to a liquid.

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Phase Diagram of Carbon Dioxide

Carbon dioxide cannot exist in the liquid state at pressures below 5.11 atm; CO_2 sublimes at normal pressures.



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